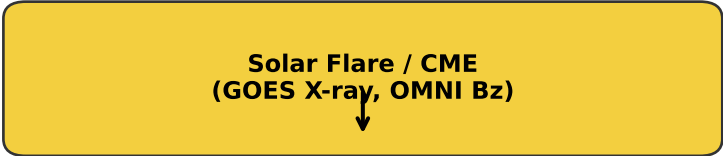
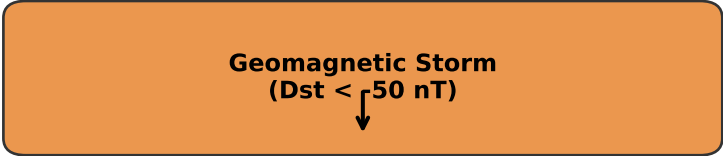


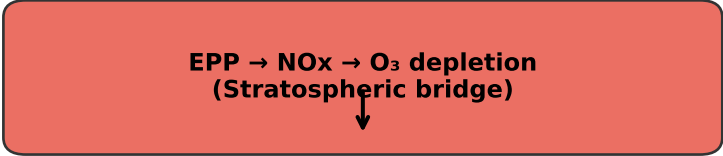
Mechanistic Chain: Solar → Stratosphere → Avalanche
with Evidence Tier Annotations



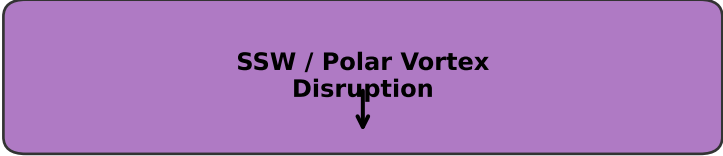
Solar data: GOES, OMNI, SDO
(SOC power-law $\alpha \approx 1.8$)



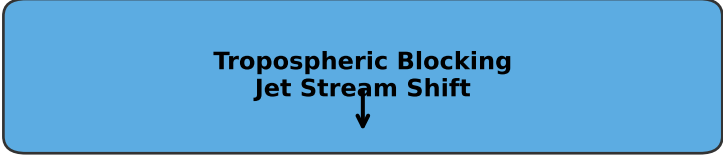
Geomag storms: 135 events
(Dst < -50 nT, declustered)



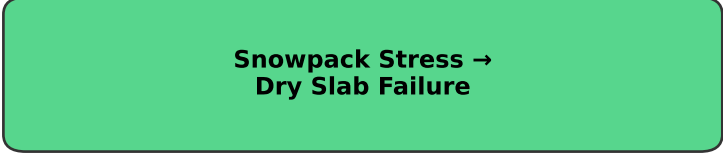
MLS NO_x, ERA5 reanalysis
(atmospheric bridge)



★ TIER 1: SSW→Dry suppression
14/15 events, all tests $p < 0.01$
Bootstrap CI excludes zero



★ TIER 1: Geomag→Dry RR=0.637
Isolated events: causal direction
POST $p < 0.0001$, PRE $p = 0.84$



★ TIER 1: Dry-specific
Wet null (RR=0.90, $p = 0.34$)
Summer null – winter only